Specifications for C Computation Tests: Entropy-Space, Doob-Transform, Schrödinger-Bridge, and Transfer-Operator Diagnostics for the Accelerated Collatz Cocycle

Research notes and implementation specification

June 9, 2026

Abstract

This document specifies a C-based computational program for testing four related formulations of the remaining Collatz obstruction: the Doob h-transform/Cramér tilt, Schrödinger bridge/entropic interpolation, deterministic transport between annealed height laws, and transfer-operator flow over residue-height states. The computations are not intended to prove Collatz. They are designed to test whether the deterministic accelerated Collatz orbit shadows, avoids, or is repelled from the annealed rare-event laws predicted by the forward height cocycle. The central empirical object is the valuation cocycle

$$b(x) = v_2(3x+1), \qquad \Delta_b = \log_2 3 - b,$$

with annealed law $p_b = 2^{-b}$ and moment transform

$$M(s) = \sum_{b \geq 1} 2^{-b} 2^{s(\log_2 3 - b)} = \frac{3^s}{2^{1+s} - 1}.$$

At the Cramér root $s = 1$, the tilted law is

$$p_b^{(1)} = p_b 2^{\Delta_b} = 3 \cdot 4^{-b} = \frac{3}{4}\left(\frac{1}{4}\right)^{b-1}.$$

The tests below are designed to determine how critical excursions, post-exit floors, and ordinary orbit windows sit relative to this tilted law and to the associated transfer-operator spectra.

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1 Executive Summary

The program has converged on a single analytic obstruction: the annealed height law is explicit and correctly predicts rare-event statistics, while the needed theorem is quenched and pointwise. In computational terms, the question becomes:

> Can a deterministic positive integer orbit repeatedly shadow the annealed rare-event flow, or does doing so force a finite spectral/character obstruction already ruled out by the finite certificate machinery?

The present specification builds computational tests around four formulations:

F1. Doob h-transform / Cramér tilt. Critical excursions should have empirical valuation law close to $p_b^{(1)} = 3 \cdot 4^{-b}$ in the bulk. Post-exit segments should fail to reacquire this law if the taboo mechanism is real.

F2. Schrödinger bridge / entropic interpolation. A critical excursion is a finite bridge: entry boundary layer, tilted bulk, exit boundary layer. Consecutive critical bridges should be forbidden or spectrally suppressed.

F3. Deterministic transport in entropy space. Sliding windows of valuation words trace paths in the simplex of empirical valuation laws. Ordinary windows should lie near p, high ascents near $p^{(1)}$, and post-exit windows should be separated from $p^{(1)}$.

F4. Transfer-operator flow over residue-height states. A family $\mathcal{L}_s$ over residue-height states should encode ordinary dynamics ($s = 0$), critical tilt ($s = 1$), QSD, H23a character gaps, and post-exit taboo through spectra such as $\rho(\mathcal{L}_1 \mathcal{E} \mathcal{L}_1)$.

The outputs are standardized CSV/JSON logs and summary tables. All exact logical predicates, especially criticality inequalities, must be evaluated using integer arithmetic. Floating-point arithmetic is used only for diagnostics such as entropy, pressure, and KL divergence.

2 Mathematical Definitions

2.1 Accelerated odd map and valuation word

For odd positive x, define

$$S(x) = \frac{3x+1}{2^{b(x)}}, \qquad b(x) = v_2(3x+1).$$

An orbit segment of length m has valuation word

$$w = (b_0, b_1, \ldots, b_{m-1}), \qquad b_i = b(S^i x),$$

and cumulative valuation

$$D_m = \sum_{i=0}^{m-1} b_i.$$

The deficit/excess relative to the equilibrium slope is

$$E_m = D_m - m \log_2 3.$$

In the forward height convention, define

$$\Delta_b = \log_2 3 - b, \qquad Y_m = \sum_{i=0}^{m-1} \Delta_{b_i} = m \log_2 3 - D_m = -E_m.$$

High ascents correspond to large positive Y_m.

2.2 Annealed valuation law and pressure transform

Under the Haar/annealed model,

$$p_b = \mathbb{P}(b = k) = 2^{-b}, \qquad b \geq 1.$$

The base-2 moment transform of the height increment is

$$M(s) = \mathbb{E}_p\left(2^{s\Delta_b}\right) = \sum_{b \geq 1} 2^{-b} 2^{s(\log_2 3 - b)} = \frac{3^s}{2^{1+s} - 1}.$$

The Cramér root is $s^* = 1$, since $M(1) = 1$. The tilted law at parameter s is

$$p_b^{(s)} = \frac{p_b 2^{s\Delta_b}}{M(s)}.$$

At $s = 1$,

$$p_b^{(1)} = 3 \cdot 4^{-b} = \frac{3}{4}\left(\frac{1}{4}\right)^{b-1}.$$

The binary entropy associated with the critical bulk law is

$$H_2(3/4) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} \approx 0.811278.$$

2.3 Critical excursions

Let V be an odd launch floor and P the peak reached during the forward excursion before the first return below V or before a specified stopping condition. Define the logarithmic height

$$h(V) = \log_2 \frac{P}{V}.$$

The A-critical launch set is

$$C_A = \{V : \ h(V) \geq \log_2 V - A\}.$$

Equivalently, for $A = 6$,

$$V \in C_6 \quad \Longleftrightarrow \quad V^2 \leq 2^6 P.$$

This predicate must be evaluated exactly as an integer inequality.

2.4 Empirical valuation laws

For a finite word w of length m, define its empirical valuation law

$$\mu_w(b) = \frac{1}{m}\#\{0 \le i < m : b_i = b\}.$$

Diagnostics include

$$\bar{b}(\mu) = \sum_b b\mu(b), \qquad \bar{\Delta}(\mu) = \sum_b (\log_2 3 - b)\mu(b),$$

$$H(\mu) = -\sum_b \mu(b)\log_2 \mu(b),$$

$$D_{\mathrm{KL}}(\mu\|p) = \sum_b \mu(b)\log\frac{\mu(b)}{p_b}, \qquad D_{\mathrm{KL}}(\mu\|p^{(1)}) = \sum_b \mu(b)\log\frac{\mu(b)}{p_b^{(1)}}.$$

3 C Implementation Architecture

3.1 Directory layout

Recommended structure:

```
collatz_entropy_tests/
  include/
    collatz_u128.h
    valuation_word.h
    histograms.h
    entropy_stats.h
    csv_writer.h
    operator_matrix.h
    config.h
  src/
    collatz_u128.c
    valuation_word.c
    histograms.c
    entropy_stats.c
    csv_writer.c
    operator_matrix.c
    common_cli.c
  tools/
    doob_tilt_test.c
    bridge_layers_test.c
    entropy_transport_test.c
    transfer_operator_test.c
    critical_floors_test.c
    post_exit_test.c
    pressure_tail_test.c
  scripts/
    build.sh
    run_all.sh
```

```
    summarize.py
  data/
    raw/
    summaries/
  docs/
```

3.2 Compiler and build flags

Use a strict, reproducible build:

```
cc -O3 -std=c17 -Wall -Wextra -Wshadow -Wconversion \
  -pedantic -march=native -DNDEBUG \
  -Iinclude src/*.c tools/doob_tilt_test.c -lm -o bin/doob_tilt_test
```

For debug builds:

```
cc -O0 -g -fsanitize=address,undefined -std=c17 \
  -Wall -Wextra -Wshadow -Wconversion -pedantic \
  -Iinclude src/*.c tools/doob_tilt_test.c -lm -o bin/doob_tilt_test_dbg
```

3.3 Integer arithmetic requirements

Use unsigned 128-bit arithmetic for all exact checks up to the intended bound:

`__uint128_t`.

Criticality must be checked as

$$V^2 \leq 2^A P$$

without floating point. The code must detect overflow before multiplication. If computations exceed 128 bits, either stop with a clear diagnostic or switch to a big-integer backend such as GMP. The first implementation target should remain within the safe 128-bit range.

3.4 Common command-line options

Every tool should support:

```
--start <uint64>          first odd V or x to test
--end <uint64>            exclusive upper bound
--octave-min <N>          first dyadic octave
--octave-max <N>          last dyadic octave
--A <int>                 criticality parameter, default 6
--max-steps <int>         maximum accelerated steps per orbit/excursion
--max-b <int>             histogram cap for valuation b
--window <int>            sliding window length
--out <path>              output CSV path
--summary <path>          output summary CSV/JSON path
--seed <uint64>           optional deterministic sample seed
--sample-rate <int>       test every kth odd integer, default 1
--threads <int>           optional threading
--verbose                 progress and diagnostics
```

3.5 Shared data structures

```
typedef struct {
    uint64_t start;
    uint64_t end;
    int A;
    int max_steps;
    int max_b;
    int window;
    const char *out_path;
    const char *summary_path;
} RunConfig;

typedef struct {
    uint64_t V; // launch floor
    uint64_t exit_floor; // first return below V, if any
    __uint128_t peak; // peak P
    int steps;
    int max_b;
    int *b_word; // length steps
    double height; // diagnostic only
    double amin; // log2(V^2/P), diagnostic only
    bool critical;
} ExcursionRecord;

typedef struct {
    uint64_t count;
    uint64_t *bins; // bins[1..max_b], overflow bin max_b+1
    uint64_t overflow;
} ValuationHist;
```

4 Core Algorithms

4.1 Accelerated step

For odd x, compute

$$y = 3x + 1, qquadb = v_2(y), \qquad S(x) = y/2^b.$$

The implementation must check overflow in $3x + 1$.

```
static inline int ctz_u128(__uint128_t y) {
    int c = 0;
    while ((y & 1u) == 0u) { y >>= 1; c++; }
    return c;
}

bool accelerated_step_u128(__uint128_t x, __uint128_t *next, int *b) {
    __uint128_t y = 3*x + 1;
    *b = ctz_u128(y);
    *next = y >> (*b);
    return true;
}
```

4.2 Excursion extraction

Given an odd launch floor V, iterate until the first return below V, a maximum step bound, overflow, or terminal cycle detection. Record:

- valuation word $b_0, \dots, b_{m-1}$;
- peak $P = \max_i x_i$;
- height $h = \log_2(P/V)$;
- criticality $V^2 \le 2^A P$;
- exit floor $E(V)$ if the excursion returns below V.

4.3 Exact criticality check

For $A = 6$, test

$$V^2 \le 64P.$$

General A:

$$V^2 \le 2^A P.$$

```
bool is_A_critical_u128(uint64_t V, __uint128_t P, int A) {
    __uint128_t lhs = (__uint128_t)V * (__uint128_t)V;
    __uint128_t rhs = P << A;
    return lhs <= rhs;
}
```

4.4 Histogram and KL divergence

For each word w, build a histogram over $b \in \{1, \dots, B_{\max}\}$, with overflow bin. Compute μ_w. Compare to

$$p_b = 2^{-b}, \qquad p_b^{(1)} = 3 \cdot 4^{-b}.$$

For the overflow bin, use the exact tail mass:

$$p_{>B} = 2^{-B},$$

for the annealed law, and

$$p_{>B}^{(1)} = 4^{-B},$$

for the tilted law, since $p_b^{(1)} = 3 \cdot 4^{-b}$.

5 Test Group I: Doob h-Transform and Cramér Tilt

5.1 Test I.1: Critical-excursion tilted valuation law

Goal. Confirm that the bulk valuation law of A-critical excursions converges to

$$p_b^{(1)} = 3 \cdot 4^{-b}.$$

Input. Odd launch floors $V \in [2^N, 2^{N+1})$ for octaves $N_{\min} \le N \le N_{\max}$.

Procedure.

1. Extract the first-return excursion from each V.
2. Retain only records satisfying $V^2 \le 2^A P$.
3. Optionally remove an entry boundary layer of length r and exit boundary layer of length r.
4. Build aggregate valuation histograms for full word and bulk word.
5. Compute $\widehat{p}_b$, $D_{\mathrm{KL}}(\widehat{p}\|p)$, $D_{\mathrm{KL}}(\widehat{p}\|p^{(1)})$, entropy, and mean height increment.

Output CSV.

```
octave,A,count_critical,total_words,total_symbols,
b,p_hat,p_annealed,p_tilted,kl_to_p,kl_to_tilt,entropy,mean_b,mean_delta
```

Expected signal. For critical-excursion bulk symbols,

$$\widehat{p}_b \to 3 \cdot 4^{-b}, \qquad D_{\mathrm{KL}}(\widehat{p}\|p^{(1)}) \to 0.$$

5.2 Test I.2: Height-tail law by octave

Goal. Verify octave-uniform tail

$$\mathbb{P}(h \ge H) \asymp 2^{-H},$$

and therefore

$$\#(C_A \cap [2^N, 2^{N+1})) = O_A(1).$$

Procedure. For each octave N and threshold H, count

$$\#\{V \in [2^N, 2^{N+1}) : h(V) \ge H\}.$$

Fit

$$\log_2 \mathbb{P}_N(h \ge H) \approx -H + c.$$

Output CSV.

```
octave,H,count,total,prob,log2_prob,slope_fit,intercept_fit
```

Expected signal. For large H below finite-size cutoff,

$$\log_2 \mathbb{P}_N(h \ge H) + H$$

should be approximately constant across octaves.

5.3 Test I.3: Post-exit conditional valuation law

Goal. Determine whether post-exit segments are anti-tilted away from $p^{(1)}$.

Procedure. For each $V \in C_A$ with exit floor $E(V)$:

1. Start a new segment at $E(V)$.
2. Record the first L_{post} valuations or the next first-return excursion.
3. Build the empirical law μ_{post}.
4. Compute $D_{\mathrm{KL}}(\mu_{post} \| p^{(1)})$ and empirical post pressure

$$M_{post}(1) = \sum_b \mu_{post}(b) 2^{\log_2 3 - b}.$$

Expected signal. If post-exit taboo is pressure-based, then

$$M_{post}(1) < 1$$

for the relevant initial post-exit window, or at least

$$D_{\mathrm{KL}}(\mu_{post} \| p^{(1)}) \geq c > 0.$$

6 Test Group II: Schrödinger Bridge / Entropic Interpolation

6.1 Test II.1: Entry, bulk, and exit boundary layers

Goal. Determine whether critical excursions decompose into entry layer, tilted bulk, and exit layer.

Procedure. For each critical word w of length m, choose boundary depth r and split:

$$w = w_{entry}\, w_{bulk}\, w_{exit}.$$

Compute valuation laws

$$\mu_{entry}, \quad \mu_{bulk}, \quad \mu_{exit}.$$

Compare each to p and $p^{(1)}$.

Output CSV.

```
octave,V,word_len,r,region,count_symbols,mean_b,mean_delta,
entropy,kl_to_p,kl_to_tilt,p1_freq,p2_freq,p3_freq,overflow
```

Expected signal. The bulk should be closest to $p^{(1)}$. Entry and exit layers should show systematic deviations. The earlier transient dimension effects should be visible in the entry layer.

6.2 Test II.2: Critical bridge non-concatenation

Goal. Test whether the exit marginal of a critical bridge is separated from the entrance marginal of another critical bridge.

Procedure. For each critical excursion:

1. Record exit floor $E(V)$.
2. Compute whether $E(V) \in C_A$ exactly.
3. Compute next-excursion height $h(E(V))$.
4. Define margin
$$\Delta_{post} = \log_2 E(V) - A - h(E(V)).$$

Output CSV.

```
V,peak,exit_floor,A,h_V,h_exit,amin_V,delta_post,
critical_again,next_word_len,next_peak
```

Expected signal. For $A = 6$, computations so far suggest

$$E(C_6) \cap C_6 = \varnothing.$$

A positive lower bound for Δ_{post} would be stronger.

6.3 Test II.3: Empirical bridge action

Goal. Verify that critical excursions minimize an entropy action predicted by the annealed bridge.

Definition. For word w of length m, define

$$\mathcal{A}(w) = mD_{\mathrm{KL}}(\mu_w \| p).$$

For a high ascent of height H, compare $\mathcal{A}(w)$ to $H \log 2$.

Expected signal. Critical excursions should have action near the Cramér prediction. Failed second excursions after exit should show excess action or inability to approach the bridge law.

7 Test Group III: Deterministic Transport in Entropy Space

7.1 Test III.1: Sliding-window empirical-law trajectories

Goal. View deterministic orbits as paths in the probability simplex of valuation laws.

Procedure. For each orbit and window length L_w:

1. Compute sliding-window law μ_{n,L_w}.
2. Record coordinates
$$\bar{b}, quad\bar{\Delta}, \quad H(\mu), \quad D_{\mathrm{KL}}(\mu \| p), \quad D_{\mathrm{KL}}(\mu \| p^{(1)}).$$
3. Label windows as ordinary, pre-critical, critical, post-exit, or terminal.

Output CSV.

```
x0,n,window,label,mean_b,mean_delta,entropy,
kl_to_p,kl_to_tilt,height_start,height_end,max_height
```

Expected signal. Ordinary windows cluster near p. Critical windows approach $p^{(1)}$. Post-exit windows avoid $p^{(1)}$ if taboo is real.

7.2 Test III.2: Distance to the Cramér curve

Goal. Determine whether orbit windows align with the exponential family

$$p_b^{(s)} = \frac{p_b 2^{s\Delta_b}}{M(s)}.$$

Procedure. For each empirical law μ, find

$$s^*(\mu) = \arg \min_{s\in[s_{min},s_{max}]} D_{\mathrm{KL}}(\mu \| p^{(s)}).$$

Record the minimizing s and the residual distance.

Expected signal. Critical windows should have $s^*(\mu) \approx 1$. Post-exit windows should move away from $s = 1$.

7.3 Test III.3: Entropic action budget

Goal. Test whether repeated critical events fail because the post-exit segment lacks enough entropy/height budget to return to the critical entropy manifold.

Procedure. For each critical excursion and following segment, compute cumulative action

$$\mathcal{A}_{0:n} = \sum_j D_{\mathrm{KL}}(\mu_j \| p)$$

or block action

$$|w| D_{\mathrm{KL}}(\mu_w \| p).$$

Compare against achieved height.

Expected signal. If an action barrier exists, post-exit attempted second ascents should require action greater than the available large-deviation budget.

8 Test Group IV: Transfer-Operator Flow over Residue-Height States

8.1 State space and operators

For fixed modulus Q and height strip $[0, L]$, define finite bins

$$\mathcal{X}_{Q,L} = \Omega_Q \times \{0, 1, \dots, L\}.$$

The tilted transfer operator is

$$(\mathcal{L}_s f)(a,z) = \sum_b p_b 2^{s(\log_2 3 - b)} f(T_b a, z + \log_2 3 - b),$$

with killing at boundaries. In implementation, z may be discretized into bins of width δz.

Residue characters decompose the operator into blocks

$$\mathcal{L}_s = \mathcal{L}_{s,0} \oplus \bigoplus_{\chi \neq 1} \mathcal{L}_{s,\chi}.$$

8.2 Test IV.1: Spectral gap as a function of tilt s

Goal. Determine whether H23a-type character gaps persist through the critical tilt $s = 1$.

Procedure. For each Q, L, and $s \in [0,1]$:

1. Build the finite matrix approximation to $\mathcal{L}_s$.
2. Compute principal spectral radius $\rho_0(s)$.
3. For each nontrivial character block, compute $\rho_\chi(s)$.
4. Record gap

$$\kappa_Q(s) = 1 - \max_{\chi \neq 1} \frac{\rho_\chi(s)}{\rho_0(s)}.$$

Output CSV.

```
Q,L,z_bins,s,rho0,max_rho_chi,kappa,argmax_chi,iterations,residual
```

Expected signal. A positive gap over $s \in [0,1]$ supports stability of the certificate under critical tilting.

8.3 Test IV.2: Doob-transformed operator paths

Goal. Compare actual critical excursions to paths typical under the Doob-transformed critical operator.

Procedure. Compute principal eigenfunction h_s of $\mathcal{L}_s$ and define

$$\mathcal{L}_s^h f = \frac{1}{\rho_s h_s} \mathcal{L}_s(h_s f).$$

Sample or enumerate typical paths under $\mathcal{L}_1^h$ and compare valuation histograms and boundary layers to actual critical excursions.

8.4 Test IV.3: Post-exit operator spectrum

Goal. Test the spectral form of post-exit taboo.

Object. Let $\mathcal{E}$ be an empirical or exact post-exit kernel from completed critical excursions to next launch floors/residue-height states.

Compute spectral radius of

$$\mathcal{K}_A \mathcal{E} \mathcal{K}_A$$

where $\mathcal{K}_A$ is the critical-entrance or critical-tilted operator. In the strongest case, the matrix is zero on the exact critical sector; otherwise one wants spectral radius < 1.

Output CSV.

```
Q,L,A,operator_type,rho_K,rho_KEK,max_entry,nonzero_edges,
spectral_gap,iterations,residual
```

Expected signal. For $A = 6$, existing tests suggest no immediate second critical event. The spectral test should show either zero transition or a strong gap.

8.5 Test IV.4: Dirac-to-QSD comparison

Goal. Quantify the annealed-to-quenched gap.

Procedure. Start from Dirac states corresponding to actual launch floors. Iterate the annealed killed operator and compare:

1. annealed distribution at time n;
2. QSD profile;
3. actual deterministic path state at time n.

Record discrepancies in total variation, residue Fourier modes, and height coordinate.

9 Auxiliary Tests for C_A and Height Law

9.1 Critical-floor census

Compute C_A for octaves and thresholds. Output:

```
octave,A,total_odd,count_critical,count_per_octave,
min_V,max_V,min_amin,max_height,mean_word_len,mean_runs
```

9.2 Valuation-word grammar diagnostics

For each critical word, record:

- total word length;
- number of $b = 1$ runs;
- longest $b = 1$ run;
- drop-size counts for $b = 2, 3, 4, \ldots$;
- entry/bulk/exit layer histograms.

This verifies that the target is not a finite grammar.

9.3 Pressure fit from observed words

For each octave and criticality threshold, estimate empirical transform

$$\widehat{M}_N(s) = \sum_b \widehat{p}_{b,N} 2^{s(\log_2 3 - b)}.$$

Compare with

$$M(s) = \frac{3^s}{2^{1+s} - 1}.$$

10 Exactness, Validation, and Failure Modes

10.1 Exact predicates versus diagnostics

The following must be exact:

- accelerated step for all tested x;
- $b = v_2(3x + 1)$;
- peak P within the integer range;
- criticality $V^2 \leq 2^A P$;
- first return below launch floor;
- cycle/terminal detection if used.

The following may be floating-point diagnostics:

- $h = \log_2(P/V)$;
- $\text{amin} = \log_2(V^2/P)$;
- entropy and KL divergence;
- spectral radii and eigenvectors;
- pressure fits.

10.2 Validation tests

Every executable should include a small deterministic test suite:

1. Check known short orbits entering 1.
2. Check $v_2(3x + 1)$ for hand-computable values.
3. Check exact criticality predicate against floating diagnostic for small values.
4. Check histograms sum to word length.
5. Check $M(1) = 1$ numerically.
6. Check $\sum_{b=1}^{B} 3 \cdot 4^{-b} + 4^{-B} = 1$ for overflow tail.

10.3 Interpretation failure modes

The computations may falsify specific hypotheses:

- If critical bulk words do not approach $p^{(1)}$, the Cramér-tilt model is incomplete.
- If post-exit windows also approach $p^{(1)}$, the taboo mechanism is not explained by anti-tilt.
- If $\rho(\mathcal{K}_A \mathcal{E} \mathcal{K}_A) \geq 1$, the spectral taboo form fails.
- If character gaps collapse near $s = 1$, H23a is not stable under critical tilt.
- If Dirac-to-QSD discrepancy remains uncontrolled, the annealed-to-quenched gap remains exactly where expected.

11 Output File Formats

11.1 Raw excursion records

```
V,octave,A,critical,steps,peak,exit_floor,height,amin,
word_len,max_b,b_word_hash,b1_runs,max_b1_run,overflow_flag
```

The full word may be optionally stored in a separate compressed text/binary file keyed by `b_word_hash`.

11.2 Histogram summaries

```
group_id,label,octave,A,region,count_words,count_symbols,b,count_b,
p_hat,p_annealed,p_tilted
```

11.3 Entropy-space windows

```
x0,n,window,label,mean_b,mean_delta,entropy,
kl_to_p,kl_to_tilt,best_s,kl_to_exp_family,height_start,height_end
```

11.4 Spectral summaries

```
Q,L,z_bins,s,block,rho,residual,iterations,
rho0,max_rho_chi,kappa,argmax_chi
```

12 Milestones

Milestone 1: Exact core and critical census

- Implement accelerated step, excursion extraction, exact criticality.
- Reproduce known C_6 counts over a small range.
- Produce octave count table for $A = 4, 5, 6, 7, 8$.

Milestone 2: Doob/Cramér diagnostics

- Compute critical bulk valuation histograms.
- Verify convergence toward $p_b^{(1)} = 3 \cdot 4^{-b}$.
- Fit height-tail law $\mathbb{P}(h \geq H) \asymp 2^{-H}$.

Milestone 3: Bridge and post-exit diagnostics

- Split critical words into entry/bulk/exit layers.
- Measure post-exit law and $M_{post}(1)$.
- Test $E(C_6) \cap C_6 = \varnothing$ at increased scale.

Milestone 4: Entropy-space transport

- Generate sliding-window entropy trajectories.
- Fit best exponential-family parameter s.
- Test whether post-exit windows avoid neighborhoods of $s = 1$.

Milestone 5: Transfer-operator flow

- Build finite residue-height operators $\mathcal{L}_s$.
- Compute character gaps over $s \in [0, 1]$.
- Construct empirical/exact post-exit kernel $\mathcal{E}$ and compute $\rho(\mathcal{K}_A \mathcal{E} \mathcal{K}_A)$.

13 Deliverables

Each round should produce:

1. source code and build log;
2. raw CSV files;
3. summary CSV/JSON files;
4. a short report with plots and interpretation;
5. exact command lines sufficient to reproduce the run;
6. a manifest of tested ranges and overflow exclusions.

14 Conclusion

The computation program should test a single organizing hypothesis from four directions:

> A deterministic branch that shadows the annealed rare-event flow for too long must force a finite spectral/character obstruction, or else reveal a new post-exit taboo mechanism in the height cocycle.

The Doob/Cramér tests identify the local rare-event law. The Schrödinger-bridge tests isolate entry and exit boundary layers. The entropy-transport tests track deterministic empirical laws in information-geometric coordinates. The transfer-operator tests connect the same phenomenon to H23a, QSD, and post-exit spectra. Together they are designed to distinguish three possibilities: a provable taboo transition, a finite spectral obstruction, or a genuinely new quenched equidistribution barrier.